

Functional Requirement Specification

Document Control Information

Document Information	Details
System Name	CutWise IMS – Inventory and Scrap Management Subsystem
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Prepared By	Josef G. Adalim – Project Manager
Reviewed By	Bonales, Paraiso, Sanchez
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Revision History

Version	Date	Description of Changes	Author
0.1	May 2026	Initial draft	Josef G. Adalim
1.0	May 2026	First Completed draft	Josef G. Adalim
2.0	May 2026	Revised FRS based on stakeholder and OTTO IT feedback; added role-based access, audit trail logging, report export functionality, user permissions, and integration updates	Josef G. Adalim

1. Introduction

1.1 Purpose

This document defines the functional requirements of the Inventory and Scrap Management Subsystem for the CutWise IMS. It serves as a guide for system development, testing, and validation.

1.2 Scope

The system covers:

- Tracking of raw materials (leather and supplies)
- Scrap management and conversion
- Inventory monitoring and updates

Excluded:

- Sales transaction processing
- Leather cutting computation
- Delivery tracking

1.3 System Overview

The Inventory and Scrap Management Subsystem is designed to manage raw materials and scrap in real time. It receives data from the Sales & Cutting subsystem and ensures proper classification, tracking, and reporting of materials.

1.4 References

- OTTO Shoes System Proposal (CutWise IMS)
- Capstone Project Guidelines
- Lecture Notes (System Analysis & Design)

1.5 Definitions and Acronyms

IMS	Inventory Management System
Scrap	Unusable leftover leather converted to kg
Off-cut	Remaining material after cutting
Real-time	Immediate update without delay

2. System Context

2.1 Business Context

The system supports inventory operations of OTTO Shoes Manufacturing by ensuring accurate tracking of materials and reducing waste.

2.2 User Characteristics

User	Description
Inventory Clerk	Encodes and updates inventory
Supervisor / Co-Admin	Monitors reports and inventory status
Admin	Manages system settings and users

2.2.1 User Roles and Permissions

Module / Feature	Inventory Clerk	Supervisor / Co-Admin	Admin
Add Material	✓	✓	✓
Update Material	✓	✓	✓
View Inventory	✓	✓	✓
Record Scrap	✓	✓	✓
View Scrap Inventory	✓	✓	✓
View Scrap Profit	✗	✓	✓
Generate Reports	✗	✓	✓
Export Reports	✗	✓	✓
Configure Settings	✗	✗	✓
User Management	✗	✗	✓
View Audit Trail	✗	✗	✓

2.3 Constraints

- Web-based system only
- Requires internet connection
- Must support PHP/NodeJS stack

2.4 Assumptions and Dependencies

- Sales subsystem provides correct material usage
- Users are trained to input correct data
- System depends on centralized database

3. Functional Requirements

3.1 Inventory Management

FR-001: Add Material

Description:

The system shall allow users to add new raw materials, including details such as material name, type, size, quantity, unit cost, and supplier.

Priority: High

Source: Stakeholder Requirement

Rationale:

To ensure accurate recording and availability of materials in the inventory system.

Acceptance Criteria:

- User can input all required fields
- System validates required fields before saving
- New material appears in inventory list
- System prevents duplicate entries

Dependencies:

- Supplier Reference Module

Status: Draft

FR-002: Update Material

Description:

The system shall allow users to edit and update existing material information.

Priority: High

Source: Stakeholder Requirement

Rationale:

To maintain accurate and up-to-date inventory records.

Acceptance Criteria:

- User can modify material details
- Changes are saved successfully
- Updated data reflects immediately in inventory

Dependencies:

FR-001

Status: Draft

FR-003: View Inventory

Description:

The system shall display all materials with details such as material name, type, size, quantity, unit cost, and status.

Priority: High

Source: System Design

Rationale:

To provide users with a clear and organized view of inventory data.

Acceptance Criteria:

- Inventory list loads successfully
- All material details are displayed correctly

- Data is updated in real time

Dependencies:

Inventory Database

Status: Draft

FR-004: Real-Time Stock Update**Description:**

The system shall automatically update inventory levels when materials are added or used in production.

Priority: High

Source: Stakeholder Requirement

Rationale:

To ensure accurate tracking and prevent discrepancies in stock levels.

Acceptance Criteria:

- Stock updates immediately after transactions
- Inventory reflects correct quantities
- No manual refresh is required

Dependencies:

- Sales & Cutting Subsystem

Status: Draft

FR-005: Low Stock Alert**Description:**

The system shall notify users when stock levels fall below a predefined threshold.

Priority: Medium

Source: Business Requirement

Rationale:

To prevent stock shortages and ensure timely restocking.

Acceptance Criteria:

- Alert is triggered when stock is below threshold
- Notification appears in the dashboard notification panel
- Optional email notification is sent to the Supervisor / Co-Admin
- Threshold value can be configured

Dependencies:

- FR-017 (Set Low Stock Threshold)

Status: Draft

3.2 Scrap Management

FR-006: Record Scrap**Description:**

The system shall record leftover materials received from the Sales & Cutting subsystem.

Priority: High

Source: Integration Requirement

Rationale:

To ensure proper tracking and utilization of leftover materials.

Acceptance Criteria:

- System receives off-cut data
- Scrap is recorded correctly
- Data appears in scrap inventory list

Dependencies:

Sales & Cutting Subsystem

Status: Draft

FR-007: Convert Scrap to KG**Description:**

The system shall convert unusable materials into scrap inventory measured in kilograms.

Priority: High

Source: Stakeholder Requirement

Rationale:

To standardize scrap measurement and support resale tracking.

Acceptance Criteria:

- Scrap is converted into kilograms automatically
- Weight is accurately recorded
- Scrap value is calculated correctly

Dependencies:

- FR-006

Status: Draft

FR-008: View Scrap Inventory**Description:**

The system shall display scrap data including material type, weight, value, and status to authorized users.

Priority: High

Source: System Design

Rationale:

To provide visibility of scrap materials and their availability.

Acceptance Criteria:

- Scrap inventory loads successfully
- All data fields are accurate
- Status (available/sold) is displayed
- Only authorized users can access the module

Dependencies:

- Scrap Database
- User Authentication Module

Status: Draft

FR-009: Record Scrap Sale

Description:

The system shall allow users to record scrap sales and update inventory accordingly.

Priority: High

Source: Business Requirement

Rationale:

To track scrap sales and generate additional revenue.

Acceptance Criteria:

- User can input scrap sale details
- System calculates total value
- Scrap inventory is updated
- Sale is recorded successfully

Dependencies:

- FR-007

Status: Draft

FR-010: Track Scrap Profit

Description:

The system shall calculate, and display profit generated from scrap sales for authorized users.

Priority: Medium

Source: Stakeholder Requirement

Rationale:

To provide insights into additional income from scrap.

Acceptance Criteria:

- Profit is calculated correctly
- Data is displayed in reports
- Updates reflect in real time
- Only Supervisor / Co-Admin and Admin users can access profit analysis

Dependencies:

- FR-009

Status: Draft

3.3 Supplier Reference

FR-011: Add Supplier

Description:

The system shall allow users to add supplier details including name, contact information, and materials provided.

Priority: Medium

Source: Stakeholder Requirement

Rationale:

To organize supplier information for efficient inventory sourcing.

Acceptance Criteria:

- User can input supplier details
- Required fields are validated
- Supplier appears in the list

Dependencies: None

Status: Draft

FR-012: View Supplier

Description:

The system shall display supplier information and materials provided.

Priority: Medium

Source: Stakeholder Requirement

Rationale:

To provide easy access to supplier information.

Acceptance Criteria:

- Supplier list loads successfully
- Information is accurate and complete
- User can view supplier details

Dependencies:

- FR-011

Status: Draft

3.4 Reports

FR-013: Generate Inventory Report

Description:

The system shall generate reports on inventory status, including stock levels, material usage, and low stock items.

Priority: High

Source: Stakeholder Requirement

Rationale:

To support monitoring and operational decision-making.

Acceptance Criteria:

- Report displays material details
- Low stock items are highlighted
- Data reflects real-time inventory
- Authorized users can download reports in PDF or spreadsheet format

Dependencies:

- FR-004

Status: Draft

FR-014: Generate Scrap Report

Description:

The system shall generate downloadable reports on scrap quantity, sales, and remaining inventory.

Priority: High

Source: Stakeholder Requirement

Rationale:

To monitor scrap performance and inventory status.

Acceptance Criteria:

- Scrap data is displayed correctly
- Sales and remaining values are accurate
- Report is accessible in dashboard
- Authorized users can download reports in PDF or spreadsheet format

Dependencies:

- FR-009

Status: Draft

FR-015: Profit Analysis**Description:**

The system shall display trends in scrap profit over time.

Priority: Medium

Source: Business Requirement

Rationale:

To analyze financial performance from scrap sales.

Acceptance Criteria:

- Profit trends are visualized
- Data is accurate
- Updates dynamically

Dependencies:

- FR-010

Status: Draft

FR-019: Export Reports**Description:**

The system shall allow authorized users to download inventory and scrap reports in PDF or spreadsheet format.

Priority: Medium

Source: Stakeholder Requirement

Rationale:

To provide accessible and printable reports for monitoring and documentation purposes.

Acceptance Criteria:

- Users can export reports successfully
- Reports can be downloaded in PDF format
- Reports can be downloaded in spreadsheet format
- Only authorized users can export reports

Dependencies:

- FR-013

- FR-014

Status: Draft

3.5 Settings

FR-016: Configure System Settings

Description:

The system shall allow Admin users to configure company name, currency, and units.

Priority: Medium

Source: OTTO IT Department Requirement

Rationale:

To adapt the system to business requirements.

Acceptance Criteria:

- Settings can be updated
- Changes are saved successfully
- System reflects updated configuration

Dependencies: None

Status: Draft

FR-017: Set Low Stock Threshold

Description:

The system shall allow Admin users to define stock alert levels.

Priority: High

Source: Business Requirement

Rationale:

To enable proactive inventory management.

Acceptance Criteria:

- User can set threshold value
- System triggers alert when threshold is reached
- Value is configurable

Dependencies:

- FR-005

Status: Draft

3.6 Audit Trail and Monitoring

FR-018: Audit Trail Logging

Description:

The system shall record and monitor user activities such as adding, updating, deleting, and modifying inventory and scrap records.

Priority: High

Source: OTTO IT Department Requirement

Rationale:

To improve accountability, monitoring, and security of system transactions.

Acceptance Criteria:

- System records user actions automatically.
- Logs include:
 - User
 - Action performed
 - Affected record
 - Date and time
- Authorized User can view audit logs

Dependencies:

- User Management Module

Status: Draft

3.7 User Authentication and Access Control**FR-020: User Authentication****Description:**

The system shall authenticate users using registered usernames and passwords before accessing the system.

Priority: High

Source: Security Requirement

Rationale:

To ensure that only authorized users can access system functionalities.

Acceptance Criteria:

- Users can log in using valid credentials
- Invalid credentials display error messages
- System restricts unauthorized access
- User roles determine accessible modules and features

Dependencies:

- User Management Module

Status: Draft

FR-021: User Management**Description:**

The system shall allow Admin users to create, update, deactivate, and manage user accounts and roles.

Priority: High

Source: OTTO IT Department Requirement

Rationale:

To manage user access and maintain system security.

Acceptance Criteria:

- Admin can create user accounts
- Admin can assign user roles
- Admin can update user information
- Admin can deactivate user accounts
- System saves changes successfully

Dependencies:

- FR-020

Status: Draft

4. External Interface Requirements

4.1 User Interfaces

- Web-based dashboard
- Table views and charts
- Forms for data entry

4.2 Software Interfaces

- Integration with Sales & Cutting subsystem
- Shared database
- The subsystem shall receive inventory usage and off-cut data from the Sales & Cutting subsystem through API-based integration.
- The subsystem shall share inventory and scrap data with the centralized Dashboard and Reporting module.

4.3 Hardware Interfaces

- Desktop / Laptop
- Optional mobile browser

4.4 Communication Interfaces

- HTTP/HTTPS
- REST API (for integration)

5. Data Requirements

5.1 Data Entities

- Materials
- Scrap
- Suppliers
- Reports
- Users
- Audit Logs

5.2 Data Relationships

- Materials → Scrap
- Suppliers → Materials
- Scrap → Sales

5.3 Data Validation

- Quantity must be numeric
- Price must be positive
- Required fields cannot be empty

6. Appendices

6.1 Wireframes and Mockups

- Inventory Management UI
- Scrap Management UI
- Dashboard

6.2 Business Rules

- Scrap is measured in kilograms
- Off-cuts are classified as reusable or scrap
- Low stock triggers alerts

6.3 Use Case Scenarios

Use Case ID	Use Case Name	Actor	Description	Main Flow	Expected Result
UC-01	Add New Material	Staff / Admin	Adds a new raw material into the system.	1. Open Inventory 2. Click Add 3. Input details 4. Save 5. Validate	Material added successfully
UC-02	Update Inventory After Usage	System (Sales)	Updates stock after material usage.	1. Sales processed 2. Compute usage 3. Deduct stock 4. Update	Inventory updated accurately
UC-03	Record Scrap from Off-Cuts	System / Staff	Records leftover materials as scrap.	1. Receive data 2. Classify 3. Convert to kg 4. Save	Scrap recorded

UC-04	Sell Scrap Materials	Staff	Records scrap sale.	1. Select scrap 2. Record sale 3. Input qty & price 4. Update	Sale recorded and inventory updated
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